Harivijay R Gunaseelan

+1 (226) 752-7014 | Email: hrathina@uwaterloo.ca | Profile: https://28harivijay.github.io/
LinkedIn: linkedin.com/in/harivijay-rg | Github: github.com/28harivijay

Skills

Languages: Python, C++, Embedded C++ (Arduino), MATLAB, JavaScript, C#, HTML, CSS,
AI/ML Tools: LangChain, FAISS, HuggingFace Embeddings, Scikit-learn, Pandas, RAG Pipelines, Groq API
Backend & APIs: FastAPI, Django, REST APIs, Webhooks, CORS, Cloudflare Tunnel
Hardware & Embedded: ESP32, I2C , Arduino Nano, Biomedical Sensors (SpO_2, Accelerometer)
Tools: Git, GitHub, VS Code, Render, Vercel, React, Arduino IDE, Unity, Bootstrap, CI/CD pipelines

Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science (BASc), Computer Engineering | 2025 - Present

- **Coursework:** Digital Computers, Numerical Methods, C++ (DSA, OOP), Embedded Systems (Arduino, sensors)

Work Experience

Accessibility Review Team Member — University of Waterloo
Waterloo, ON (Remote, Project-Based) June 2026 - August 2026

- Audited **HTML source code** across **3 UW PD courses** against **WCAG accessibility standards**, identifying and documenting content quality issues impacting assistive technology users across multiple course modules.
- Produced **structured compliance reports** with cited WCAG criteria for each finding, delivering clear, verifiable, and actionable documentation that enabled stakeholders to resolve accessibility gaps efficiently.
- Aligned classification and validation standards through recurring team syncs and created structured, screen-reader-friendly transcripts translating visual course content into accessible text.

Projects

Automated Job Recommendation Platform
View Repository | *Personal Project* | View in Portfolio March 2026 - April 2026

- Built an **ML-based job recommendation system** using **Python, Pandas, and Scikit-learn** trained on **500+ real postings** across **20+ companies**, automatically matching user skillsets to relevant roles.
- Trained a **Random Forest classifier** for role matching, applied **regression** for salary prediction and **K-Means clustering** for targeting; validated performance via **80/20 train-test splitting** and accuracy scoring.
- Surfaced **top-10 job matches** with salary estimates through a **Django + Bootstrap** dashboard in under **5 seconds**.

DocuMind — AI Document Q&A Platform
View Repository | *Personal Project* | Project Link | View in Portfolio April 2026 - May 2026

- Built a **full-stack RAG application** (**React, Tailwind CSS, FastAPI**) supporting **PDF, DOCX, and XLSX** uploads, enabling natural-language queries over multi-page documents without manual searching.
- Designed an end-to-end retrieval pipeline (ingestion → chunking → **HuggingFace** embeddings → **FAISS** vector search → **Groq/LangChain** generation) returning context-aware answers grounded in uploaded document content.
- Configured recursive text splitting with **500-token chunks** and **50-token overlap** to preserve context across chunk boundaries and improve retrieval consistency for multi-paragraph document-based queries.

PRISM — Patient Response Insight & Surveillance Monitor
View Repository | *Hardware Lead* | View in Portfolio Nov 2025 - Dec 2025

- Developed a wearable **HID (Hospital Induced Delirium) detection system** using **ESP32** and biomedical sensors, reducing nursing monitoring workload by **20%** and enabling significantly faster clinical response times.
- Programmed **real-time embedded firmware in C++ (Arduino)** for continuous biomedical data acquisition, optimizing buffering and I2C communication handling to minimize sensor data loss under live operating conditions.
- Benchmarked **ESP32** against Arduino Nano R4 Minima across key data throughput metrics, achieving **15× faster** handling for SpO_2 and accelerometer sensor streams in a resource-constrained wearable environment.

Achievements & Certifications

- **FreeCodeCamp Python Developer Certification**: 300-hour certification in Python fundamentals, OOP & DSA.
- **MATLAB Onramp** (MathWorks): Certification in MATLAB syntax, matrix operations, data import and plotting